

says Chinmay Anand Misra, certified LabVIEW developer, technical marketing engineer, National Instruments. He adds, "Because of these multiple redundant components, the instruments take up a lot of space, increase the cost of the set-up, and are in general cumbersome to work with if multiple of them are used at a time."

An upcoming trend, to solve the above issue, is integration of diverse instrumentation equipment, such as oscilloscope, function generator, multimeter and power supply in a single box using modular packages and providing a software for test and measurement. A user can interact with the instrument via software applications that run on PCs or iPads. This opens up new possibilities for design engineers, entrepreneurs and even academicians.

National Instruments recently announced its VirtualBench that has a lot of functionality within it. It incorporates a mixed-signal oscilloscope (2ch analogue, 32ch digital) having 100MHz bandwidth, 1G/s sampling rate and 8-bit), function generator (20MHz sine, 14-bit, 125 MS/s), digital multimeter (five-and-a-half digit, 300V and 10A max), DC power supply (+6V @ 1A, +25V @ 0.5A, -25V @ 0.5A) and digital

Five of the Latest Modular Instruments		
Product	Company	Best Features
 US309A 8-bit PCIe Gen2 high-speed digitiser	Keysight	8-bit resolution and a DC to 500MHz frequency range - Up to 2GB DDR3 on-board memory and acquisition control using a Xilinx Virtex-6 FPGA 2 channels up to 2GS/s or 8 channels up to 1GS/s - Includes a Xilinx Virtex-6 FPGA, allowing implementation of custom real-time processing algorithms using the available US340A FPGA development kit
 EMX-1434	VTI Instruments	4-channel, 24-bit arbitrary waveform generator with dual-channel 1MHz counter/timers and four channels of digital I/O that plug into the PXI Express bus - Achieves a sampling rate of 204.8k/s per channel and provides a spurious-free dynamic range of -115dB - Output modes include sine, burst sine, chirp, burst random, and continuous random
 BI-40	Brilliant Instruments	PXI-based clock generator card, available in 2.1GHz and 4.1GHz versions, generates square-wave clock signals with 20-digit resolution
 PXIe-4139	National Instruments	- A precision system source measure unit (SMU) that performs sample acquisitions at speeds of up to 1.8M/s - Up to 17 SMU channels in one 4U, 48.3cm (19-inch) PXI chassis, minimising overall test-system footprint
 SMX-3276 3U PXI Express multiplexer	VTI Instruments	- Switches up to 300V DC or AC and 2A 152 channels for scanning multiple points to a common bus in 1-wire, 2-wire, or 4-wire configurations

I/O (8ch LVTTTL). Though it aims at simplifying circuit debugging and

validation with a diverse functionality, it costs US \$2000.

PXI modular hardware architecture maximises throughput

PXI is being used for testing wireless, multi-format devices. With today's advanced PXI modules, cellular devices can be tested for standards such as TD-SCDMA, W-CDMA, cdma2000, 1xEV-DO, GSM/EDGE-Evo and LTE-Advanced. Even devices with other connectivity formats, such as 802.11ac, Bluetooth, WiMAX and multi-satellite GNSS for navigation, can be tested.

Testing MIMO RF IC. For bring-up and design verification of MIMO RF IC chipsets used in mobile devices, PXI-based devices are equipped with 18-slot PXIe chassis powered by Core i5 processors. LitePoint's zSeries MIMO includes a perfect combination of hardware and software features for testing the latest generation of MIMO RF IC chipsets. It has powerful software for signal analysis, creating waveforms and performing tests such as SISO, direct mapped MIMO, asymmetric MIMO and true MIMO testing.

Multi-device testing. Keysight's E6640A EXIM wireless test set has high-density PXI transmit and receive (TRX) modules that allow four transmit and receive channels within each unit. In sync with the latest cellular and WLAN chipsets, this test set is designed for multi-device testing with up to four TRXs.

Field-testing and DAQ. A14-slot PXI card is incorporated inside the MTS-207 test box solution from Marvin for field-testing and data acquisition systems. The MTS-207 includes a 14-slot combination, shock-mounted PXI chassis with seven 3U slots and seven 6U slots. An internal heater keeps the modules stable at sub-zero temperatures, while the modules are carefully designed or selected to meet the +65°C requirement at the high end. Internal shock absorbers cushion the PXI modules, while conformal coating protects the modules against humidity and other harsh environments. Compact PCI, closely related to PXI, is also deployed.

Because of these advancements in PXIs, the latest trend today is to use modular devices and boards to create one-box testers (OBTs).

FPGAs for more power

Another interesting development is the concept of software-defined radio (SDR). Although SDR has been around for many years, it has become really viable only in the recent years. The main idea is to configure an FPGA within a PXI instrument exactly as per required test specifications. A software-defined virtual instrument makes available raw data from the hardware to the end users so they can define their own measurements and user interfaces. A combination of general-purpose, modular hardware and software-defined virtual instrumentation enables an end user to exploit the full flexibility and performance of the measurement software.

FPGAs allow implementation of custom real-time processing algorithms. Consider Keysight's PCIe